

1 Literature Study Table

Title of Paper	Methods Used	Features	Advantages	Disadvantages	Performance Metrics	Future Scope
A Novel Approach for Crowd Analysis and Density Estimation by using ML Techniques	- MobileNet SSD, - YOLO v4 - Mask-RCNN	- Lightweight and versatile - Adaptable to real-time processing	- MobileNet SSD: Lightweight, versatile, devices with limited computational power. - YOLO v4: High precision, real-time processing - Mask-RCNN: Excellent instance segmentation, contextual	- MobileNet SSD: Reduced accuracy in high-density crowds. - YOLO v4: High computational demand. - Mask-RCNN: Slow processing, computationally heavy	- Accuracy, precision, recall and F1 score chosen for quantifying detection performance. Average Accuracy : - MobileNet SSD: 64.75% - YoloV4: 84% - Mask-RCNN: 78.5%	Real-time optimization for dense crowds
A Proposal on Stampede Detection in Real Environments	- Optical flow computation - feature extraction, in real time. - Stacking classifiers to detect stampedes	- No thresholds needed - Public dataset use	- Real-time detection - Public dataset robustness - Doesn't require predefined thresholds. - Stacking classifier enhances performance. - Public datasets ensure robustness.	- Limited performance in high-density crowds - Limited performance in high-density crowds. - Relies on optical flow techniques. - Applicable only to a small number of scenarios.	- Receiver operating characteristic (ROC) - Area under the curve (AUC) metrics - Cost of Computation	Further research for high-density crowds

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Abnormal Crowd Behavior Detection Using Motion Information Images and CNN	- Optical flow generate Motion Information Images (MIIs) - CNN classified video frames to detect abnormality	- MIIs for motion representation - Abnormality detection	- MIIs allow for better representation of crowd motion. - Achieves state-of-the-art results. - CNN architecture is simple, 3 layers.	- Computationally intensive - Limited to binary classification (normal/abnormal behavior). - MII formation takes a considerable amount of time.	UMN Dataset : MII + CNN: 99.08 : MII + ResNet-50: 98.89 PETS2009 : MII + CNN: 98.39 : MII + ResNet-50: 97.04	Explore multi-class detection
TransCrowd: Weakly-Supervised Crowd Counting with Transformers	- Weakly-supervised crowd counting model with tranformer-encoder and regression - Image Tokenization - Regression head using regression token and Global Average Pooling (GAP) method.	- Count-level annotations - Global receptive fields	- Transformers capture long-range dependencies in crowd images. - Transformers offer a global field for better context modeling. - Cheaper count-level annotations	- Requires large datasets for training. - High computational expense vs CNN	- TransCrowd - Token : (Low): MAE: 7.1, MSE: 10.7 (Medium): MAE: 33.3, MSE: 54.6 (High): MAE: 302.5, MSE: 557.4 (Overall): MAE: 58.4, MSE: 201.1 - TransCrowd - GAP : (Low): MAE: 6.7, MSE: 9.5 (Medium): MAE: 34.5, MSE: 55.8 (High):MAE: 285.9, MSE: 532.8 (Overall): MAE: 56.8, MSE: 193.6	Validate on diverse datasets.

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Machine Learning for Dense Crowd Direction Prediction Using LSTM	◦ LSTM model predicts movement trajectories based on past positions. ◦ Cone of vision concept for influence detection. ◦ "Unit Angle" quantifies these neighbors.	◦ Trajectory prediction ◦ Cone of vision concept	◦ Suitable for both structured and unstructured crowds. ◦ Retains long-term dependencies for more accurate predictions. ◦ Improves trajectory prediction for individuals in crowds.	◦ Requires careful tuning for optimal performance. ◦ Expensive requiring high-quality sequential data.	◦ Promising prediction accuracy	◦ Real-world testing required due to complexity.
Fusion of CCTV Video and Spatial Information for Automated Crowd Congestion Monitoring	◦ CCTV video integration with Crowd mobility graphs (CM graph) ◦ Trajectory is predicted with real-time monitoring. ◦ Kalman filter predictor is used to develop SORT & DeepSORT for tracking.	◦ Real-time monitoring ◦ Visualization of congestion	◦ Predictive capabilities ◦ Real-time monitoring. ◦ Effective in visualizing congestion across urban spaces.	◦ Depends on video quality & computational resources ◦ Challenges with dense crowds	◦ YoloV7-tiny + SORT: MODA (62.8%), MOTA (70.8%), Recall(72.2%), Precision(98.2%) ◦ YoloV7-tiny + DeepSORT: MODA (73.7%), MOTA (81.2%), Recall(86.2%), Precision (94.6%)	◦ Improvements needed for high-density scenarios

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A CNN-RNN Neural Network Join LSTM for Crowd Counting	◦ CNN for feature extraction & ◦ LSTM for context relevance to estimate crowd density	◦ Estimates crowd density	◦ High accuracy in dense settings ◦ Robust across densities	◦ Poor performance in sparse crowds	◦ Effective for high-density scenarios	◦ Optimization needed for sparse environments
Jhu-Crowd++: Large-Scale Crowd Counting Dataset and Benchmark Method	◦ Large-scale dataset ◦ Benchmark methods	◦ Diverse scenarios ◦ Rich annotations	◦ Diverse scenarios and environmental conditions. ◦ Rich annotations at both image and head level. ◦ Distractor images.	◦ Annotation complexity ◦ Limited diversity in distractors. ◦ Limited focus on temporal data.	◦ Robust for crowd counting	◦ Exploration of temporal data needed ◦ Real-time performance optimization ◦ Generalization to unseen scenarios ◦ Enhancing the dataset diversity and leveraging multi-camera systems.